THE QUESTION OF DEFINING OUTER SPACE

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I. Development of the Question

The problem of whether and how to define outer space, particularly where to draw its lower limit and thus to distinguish it from airspace, belongs, without any doubt, to the oldest and most discussed questions of the law of outer space and its doctrine.

As early as in 1932 Dr. Vladimir Mandl, an attorney from Plzen and at the same time an enthusiast for industrial and technological progress, who later became Professor of industrial law at the Czech Technical University of Prague, emphasized in the first monographical study on space law that the prospect of penetrating into outer space by means of rockets would open a new problem, not settled by air law which regulated only the legal regime of airspace; rockets in their capacity as means of transport going beyond dense layers of atmosphere would be based on quite a different principle than aircraft. Dr. Mandl held the view that the scope of the principle of the State territorial supremacy in airspace went far beyond the limits of aeronautics; this principle rather acknowledged the right of each State to retain sovereign power over any use of its superjacent air zone, be such use performed by aeronautics or in any other manner, including flights into outer space. As soon as a spacecraft enters a state's airspace, it will be subject to its jurisdiction. At the same time, Dr. Mandl made it abundantly clear that the adoption of the principle of sovereignty included only airspace, "l'espace atmospherique" in the sense of Art. 1 of the Paris Convention on Civil Aviation of 1919; there was no reason or intention to grant to States any rights reaching farther into space. Where airspace ends, the territorial supremacy, as mutually regulated by States both in treaties and in practice also ends.[1]

Numerous articles and attractive discussions on the issue of delimitation of outer space from airspace appeared shortly before and in the first years of space flights. During that period, it was particularly Professor John Cobb Cooper, the first Director of the Institute of Air and Space Law at McGill University in Montreal, who initiated this debate and insisted on an early solution to this problem.[2] He modified his views several times, but his final attitude formulated in his paper presented at the Sixth Colloquium on the Law of Outer Space of the IAF International Institute of Space Law in Paris, 1963, deserves to be quoted word by word. For Cooper urged

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The views expressed in this paper are those of the author and are not necessarily connected with any organization of which he is a member.

[1] V. Mandl, Das Weltraum-Recht. Ein Problem der Raumfahrt 3, 18, 31 (1932).

[2] Cooper, High Altitude Flight and National Sovereignty, Int'l L. Q. 411-418 (July 1951).

> "a new convention which will re-affirm the complete and exclusive sovereignty of a subjacent State up to the height to which normal types of aircraft may be operated, then an agreed extension of sovereignty up to the height at which an artificial satellite may be put in orbit around the earth, such convention providing, however, for the passage of peaceful spacecraft through the extended area when ascending toward or descending from free outer space above".

In specific terms he recommended a convention providing

> "complete and exclusive sovereignty of the subjacent State up to 20 or 25 miles above the earth's surface; then it would authorize the extension of national sovereignty up to 70 or 75 miles, with rights of passage for peaceful spacecraft through this extended area; . . . outer space beyond be free and not subject to national appropriation".[3]

In a similar spirit, Professor Cooper reaffirmed this position in one of his last articles dealing with the subject in which he wrote:

> "General agreement appears to exist that the upper boundary of national airspace is not below the upper limit of practical flight of aircraft requiring aerodynamic lift and using air breathing engines. This limit may be approximately 25 miles above sea level. Also, agreement seems to exist that an international rule of customary law is now effective which prevents any State from claiming sovereign territorial control in areas where orbital flight of artificial earth satellites is practical. This limit has been estimated at between 75 and 90 miles above the surface of the earth at sea level. It would appear, therefore, that there is now practical agreement that the upper level of national airspace is not less than approximately 25 miles above the earth's surface and not more than 90 miles. This question must finally be settled if the future use of 'outer space' beyond the 'airspace' is to be protected for the benefit of all States".

At the same time, however, Cooper admitted that "Certainly many States may still be unwilling to commit themselves as to a definite upper boundary of national airspace territory for military or other reasons".[4]

Since 1957, the need for and possibility of a delimitation between airspace and outer space was also defended by Andrew G. Haley, the then General Counsel of the International Astronautical Federation, who elaborated his point of view on a doctrine of Dr. Theodor von Karman, an outstanding expert in the field of aerodynamics and theory of space flight. According to Haley

> "there are two borderlines for continuous flight with aerodynamic lift—the heat barrier, which determines the maximum velocity, and the altitude barrier, which is ratio between lift and Kepler force. Between these two barriers there is a corridor of continuous flight which terminates when at [the] approximate speed of 25,000 feet per second and an altitude of about 275,000 feet the Kepler force takes over and

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[3]Cooper, The Upper Airspace Boundary Question, Proc. Sixth Colloquium on the Law of Outer Space 6 (1963).

[4]Cooper, Background of International Public Air Law, 46 Yb. Air & Space L. 26-27 (1965).

> aerodynamic lift is gone. This is a critical jurisdictional line, marking the theoretical limit of air flight, which is here termed the von Kármán line. . . The von Kármán primary jurisdictional line may eventually remain as presented above or, as a result of such developments as improved techniques of cooling and more heat-resistant materials, it may be significantly changed. But these changes will be only in the exact location of the von Kármán line, for the existence of the line is certain, and wherever it is finally drawn will be the place where 'airspace' terminates."[5]

A third example of early suggestions in the doctrine of space law concerning the definition and delimitation of outer space is embodied in the Draft Code and Rules of the David Davies Memorial Institute of International Studies of 1962. They were worked out by a Study Group headed by Professor R.Y. Jennings of Cambridge University. In this remarkable document, "airspace" was defined as the volume of space between the surface of the Earth at sea level and an altitude of 80,000 meters above it, while "outer space" was defined as space outside the airspace. In the commentary to these definitions it was recognized that at that time the lower effective limit of perigee was in the region of the altitude of 100 miles, since below that the life of the satellite was too short to be useful, and it was possible that an altitude of about 70 miles would be the limit for effective orbiting, inasmuch as below that, friction would become too great. In this document it was also confirmed that 25 miles was probably the outside limit of effective aerodynamic lift. However, three considerations were recalled, which favored a definition of airspace yielding a more extended sovereignty than 25 miles: (1) the fact that airspace begins to lose its character as a continuous medium only when a height of 50-55 miles is reached; (2) the likely range of effective control of objects from the ground; and (3) the logic of treating the frontier between airspace or outer space as being at or near orbiting altitude. In their final sentence the authors of this document admitted that any particular altitude chosen as the limit of sovereignty over the airspace might appear arbitrary and be controversial; but, for the avoidance of excessive claims and in view of the aforementioned considerations, the relatively low altitude of about 50 miles was suggested as the limit of sovereignty and the beginning of outer space.[6]

Not much later than the first suggestions concerning the delimitation between airspace and outer space, the first contributions opposing the idea of demarcation as well as the need to draw any limit of this kind also appeared. The sponsors of this position, which was soon called "functional approach", recommended regulation of different types of activities to be developed in space without any borderline between its different parts. The origin of this approach is usually attributed to two US scholars—Professor Myres S. McDougal and Leon Lipson,[7] but it was also elaborated by Professors R. Quadri

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[5]A. Haley, Space Law and Government 98-99 (1963).

[6]Draft Code and Rules of the David Davies Memorial Institute of International Studies, Draft Code of Rules on the Exploration and Uses of Outer Space in W. C. Jenks, Space Law, App. X, 419-421 (1965).

[7]McDougal and Lipson, Perspectives for a Law of Outer Space, 52 Am. J. Int'l L. 407 (1958).

of Italy and Ch. Chaumont of France[8]. Such position found able supporters among some lawyers from the Socialist countries too, particularly in the writings of Dr. Gyula Gál of Hungary who emphasized in his book on space law that

> "In accordance with the above conception of the functional character of space law, the new legal domain, other than all earlier norms of human conduct, cannot be associated with any limited space (area, zone), but only with the character of activity under regulation. . . . Its recognition is a logical necessity following from the various aspects of space law. It can be said that, though there are variuos contradictions in the relevant positive law fundamentals, this theory has been borne out during the legal progress of the past few years".[9]

Since 1959, the subject of delimitation was already listed among possible topics of the emerging space legislation to be considered by the *Ad Hoc* Committee on the Peaceful Uses of Outer Space. In its report, however, this Committee expressed the following view: "it was generally believed that the determination of precise limits for airspace and outer space did not present a legal problem calling for priority consideration at this moment".[10]

The present Committee on the Peaceful Uses of outer Space, which effectively started the space treaty-making process in 1961, maintained, at least in the first stage of its negotiation, a similar position as had been held by its *Ad Hoc* predecessor. For several succeeding years this UN body left the question of defining outer space completely aside when it considered the first draft agreements to govern the fast developing space activities.

It must, however, be recalled that only a few years later, in its Resolution 2222/XXI of December 19, 1966, by which the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies was commended for signature and ratification by States, the UN General Assembly requested the UN Committee on Outer Space "to begin at the same time the study of [the] question relative to the definition of outer space and the utilization of outer space and celestial bodies. . ."[11]. And indeed, this request was complied with in the ensuing work of the UN Committee and both of its Subcommittees.

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[8]Quadri, Droit international cosmique, 98 Recueil des Cours 505 (1959); Chaumont, Les perspectives que doit adopter le droit de l'espace, 7 Revue de droit contemporain 5 (1960).

[9]G. Gal., Space Law 106-107 (1969).

[10]Report of the Ad Hoc Committee on the Peaceful Uses of Outer Space. U.N. Doc. A/4141 at 93-94 (1959).

[11]21 U.N. GAOR, Supp. 16, at 13. U.N. Doc. A/6316 (1966).

The Legal Subcommittee discussed the question in greater detail at its following session in 1967. Consideration of the question led to divergent opinions, but it may be said in general that more or less reluctant views were held. As a matter of fact, the only positive result of this discussion was a Questionnaire addressed to the Scientific and Technical Subcommittee, inviting this body "to draw up a list of scientific criteria that could be helpful to the Legal Sub-Committee in its study relative to a definition of outer space". Furthermore, the Scientific and Technical Subcommittee was called upon to "give its views on the selection of scientific and technical criteria that might be adopted by the Legal Subcommittee, and to indicate, on scientific and technical grounds, the advantages and disadvantages of each of them in relation to the possibility of a definition which would be valid for the long-term future. . ."[12].

This way of approaching the question of defining outer space—which was after all a problem of a legal nature—must have proved to be ineffective. Indeed, after an exchange of views on this topic, the Scientific and Technical Subcommittee agreed on a position that read as follows: ". . . it is not possible at the present time to identify scientific or technical criteria which would permit a precise and lasting definition of outer space. . ."[13]. Nevertheless, the Subcommittee felt it appropriate to continue its work in this field.

For several subsequent years, this point remained on the agenda of the Legal Subcommittee, but a thorough consideration of all aspects involved was repeatedly postponed. As it was recorded in one of the Subcommittee reports of this period on this item, "owing to lack of time, it had not been considered in any detail",[14] despite the fact that—as recorded in another report—"a number of useful statements were made by delegations in which they stressed the renewed importance of the subject and expressed the wish that it should receive more detailed examination at future meetings."[15]

It should be recalled, however, that in 1970 the Legal Subcommittee initiated the preparation by the UN Secretariat of a background paper on the question. Such document was to take into account "both the data provided by the study carried out by the Legal Subcommittee and the Scientific and Technical Subcommittee, and also the contributions, studies, data and documents which may be obtained from the specialized

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[12]Report of the Legal Subcommittee on the Work of Its Sixth Session, June 19-July 14, 1967, U.N. Doc. A/AC. 105/37, at 8 (1967).

[13]22 U.N. GAOR, Annexes, Agenda Item No. 32 at para. 36, U.N. Doc. A/6804 (1967).

[14]Report of the Legal Subcommittee on the Work of its Sixteenth Session, March 14-April 8, 1977, U.N. Doc. A/AC.105/196, at 9 (1977).

[15]Report of the Legal Subcommittee on the Work of its Fifteenth Session, May 3-29, 1976, U.N. Doc. A/AC. 105/171 (1976), at 8.

agencies concerned and such other international and national organizations and institutions which are interested in the subject as may be determined by the Committee on the Peaceful Uses of Outer Space".[16] Indeed this paper, called The Question of the Definition and/or the Delimitation of Outer Space,[17] later on completed by an Addendum,[18] has offered an excellent review of opinions expressed on the problem both in intergovernmental and nongovernmental bodies and included a comprehensive analysis of all aspects involved.

In this connection, it should also be remembered that among the nongovernmental organizations, it was the International Academy of Astronautics (IAA) and the International Institute of Space Law (IISL), both of them permanent institutions of the International Astronautical Federation, which favorably responded to the need of studying both the scientific and legal aspects of the problem. The Scientific-Legal Liaison Committee, a joint body of the IAA and the IISL, discussed different aspects of the definition of outer space at two of its meetings, held respectively at the Belgrade and New York Congresses of the IAF in 1967 and 1968.[19]

Nevertheless, it was not until the recent period in the work of the UN Legal Subcommittee that new interest in a thorough consideration of the problem appeared. In our opinion, this development was stimulated by several factors of a completely different nature.

The first of these factors might be characterized as an inevitable consequence of the growth of space legislation which completed up to 1975 four international treaties of universal interest, governing different kinds of space activities. When applying the space legal instruments, in which such terms as "outer space", "space objects", "space activities" etc., are repeatedly used, States, both performing and not performing space activities, cannot live forever without knowledge of the proper meaning of outer space in precise legal terms and, particularly, of its exact beginning. For the notion of space activities itself can be defined only on the basis of the definition of outer space.[20] Of course, this factor alone, being of an intrinsic quality, would hardly be sufficient to reverse the general reluctance of giving an answer to the question which has survived for decades.

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[16]U.N. GAOR, Twenty-fourth Session, Supp. 21, Annex III, para. 13B, U.N. Doc. A/7621 (1969).

[17]U.N. Doc. A/AC. 105/C 2/7 (1970).

[18]U.N. Doc. A/AC. 105/C. 2/7/Add. 1 (1977).

[19]See Proc. Tenth Colloquium on the Law of Outer Space 267-279 (1969); Proc., Eleventh Colloquium on the Law of Outer Space 371-395 (1968).

[20]A. Piradov (ed.) Mezhdunarodnoe Kosmitcheskoye pravo 5 (1974).

In addition, however, other factors have emerged or become more impending in recent years. One of these was the ever-growing number of space objects launched into orbits around the earth and beyond them, as well as the prospect of establishing large orbiting space systems of a multipurpose character in the near future. Moreover, a new generation of space vehicles, such as the Space Shuttle, has recently initiated a study of all problems involved, including the scope of application of space law agreements to different phases of the flight of such aerospace transport vehicles.

Furthermore, the prospects of establishing solar power satellites collecting energy in outer space, converting such energy to microwave beams and transmitting it from satellites to earth[21], also raised the question of defining outer space in the particular context. For such beams will have to pass not only through the international area of outer space, governed by the principle of freedom of exploration and peaceful use of outer space for all nations but also through the zone of airspace, with due respect for the principle of complete and exclusive sovereignty of subjacent States over their relevant zones.[22]

Without any doubt, a strong impetus to a renewed consideration of the problem of defining outer space was also given by another issue which has recently emerged on the international level, namely, the legal status of the geostationary orbit. This issue appeared in connection with claims of a group of countries to the parts of the geostationary orbit superjacent to their national territory. As we know, such claims were advanced particularly in the Bogota Declaration of 1976, signed by eight equatorial countries and published on the eve of the International Telecommunication Union's World Administrative Radio Conference for Broadcasting Satellite which was convened for January 1977 to Geneva. They were also brought before the UN Committee on Outer Space and its Legal Subcommittee.[23]

As a result, the UN General Assembly, in its Resolution 196/XXXII of December 20, 1977, amplified the original wording of the item, by introducing a new formula spelled out as "Questions relating to the definition and/or delimitation of outer space and outer space activities, also bearing in mind questions relating to the geostationary orbit". It was in this manner that the question of defining outer space received its present shape which now deserves, without any doubt, new attention and careful consideration by the world community of space lawyers.

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[21]Gorove, Legal Aspects of Solar Power Satellites: Focus on Microwave Exposure Standards, Paper submitted to the Twenty-Second Colloquium on the Law of Outer Space of the International Institute of Space Law, Munich, Germany, at 1 (1979).

[22]Haanappel, Definition of Outer Space and Outer Space Activities, Proc. Twentieth Colloquium on the Law of Outer Space 53 (1977).

[23]Report of the Comm. on the Peaceful Uses of Outer Space, U.N.GAOR, 33rd Sess., Supp. 20 at 10, para. 40, U.N. Doc. A/33/20 (1978); Report of the Legal Subcomm. on the Work of Its Seventeenth Session, March 13-April 7, 1978, at 10, para. 40, U.N. Doc. A/AC.105/218 (1978).

II. *Analysis of Some Aspects of the Question*

a) *Space Law Instruments and the Definition of Outer Space*

Does present space legislation require an exact answer to the question of where outer space begins? Do the up-to-date space agreements provide a satisfactory basis for such definition? It is our conviction that both questions may be answered affirmatively.

In the first place, such conclusion may be drawn from the main space law instrument, namely the Space Treaty of 1967. Certainly, most of its provisions have been spelled out as principles governing space activities as performed from their very outset until the end rather than as provisions bolstering the legal status of outer space as such. Nevertheless, there are some other important provisions in the Space Treaty, the realm of which requires a clear idea about what is to be considered as outer space for the purposes of cosmonautics and where it begins, with all legal consequences resulting from such determination. The gist of this category of provisions lies in Article I of the Space Treaty in which the fundamental principle of freedom of exploration and use of outer space, including the moon and other celestial bodies, by all States has been incorporated. In a similar way, the principle of nonappropriation of outer space, as included in Article II of the Treaty, requires the same kind of delimitation of its scope. While outer space is governed by the common interests of all nations and shall remain free, the lower part of space superjacent to the territory of any State is subject to the latter's complete and exclusive sovereignty. Both fundamental provisions bear clearly a territorial character and cannot be correctly interpreted and applied without a precise delimitation of the two legally different spaces.

As might be expected, other principles of the Space Treaty also require a precise definition of outer space, since any violation of them by acts occurring in outer space would be connected with some legal consequences. Thus according to Article VI, States Parties to the Space Treaty shall bear international responsibility for national activities in outer space and for assuring that such activities are carried out in conformity with the provisions of the Treaty. According to Article IX, States Parties to the Treaty shall pursue studies of outer space, including the moon and other celestial bodies, and conduct exploration of them so as to avoid their harmful contamination. They are obliged to undertake, and also entitled to request, appropriate international consultations if they have reason to believe that a space activity or experiment in outer space would cause potentially harmful interference with activities of other States Parties in the peaceful exploration and use of outer space.

As another example, Article VIII of the Space Treaty might be recalled. According to its provision a State Party to the Treaty on whose registry an object launched into outer space is carried shall retain jurisdiction and control over such object and over any personnel thereof while in outer space or on a celestial body. Moreover, ownership of objects launched into outer space, including objects landed or constructed on a celestial body and their component parts, shall not be affected by their presence in outer space or

on a celestial body or by their return to the earth. Similarly as it has been in the case of freedom and nonappropriation of outer space, jurisdiction and control over space objects and the personnel thereof, as well as ownership of such objects and of their component parts are purely legal concepts requiring an exact definition of their meaning and application by means of a definition of outer space. For without such definition it could not be established whether and at which moment the State exercising its jurisdiction and control over a space object and the personnel thereof had to take into account that the object concerned entered the part of space in which the sovereignty of another State would apply.

The legal regime of outer space, being the proper theatre of space activities, widely differs from that of airspace in which aeronautics develop. It was just this difference that led the earlier proponents of a solution to this problem to emphasize the need of an agreement on a dividing line between the two parts of space. This may be illustrated by the view of Professor Alex Meyer of the University of Köln am Rhein (Federal Republic of Germany) who stated in his contribution to the First UN Conference on the Exploration and Peaceful Uses of Outer Space in Vienna, 1968, the following:

> ". . . it seems to be beyond any doubt that the States will not and cannot renounce the sovereignty over the air space above their land and water territories constituting a part of their sovereign territory. But if it is true, the fixation of a limit between air space and outer space will become a necessity, for only in this way can it be stated that an act of legal relevance, if it happened, did take place in the air space or in outer space. This statement is again necessary because, apart from international rules agreed upon, in the air space above the territories of the States—which is, as mentioned, subject to their sovereignty—the laws of the subjacent State are applicable, whereas in the free outer space acts of legal relevance on board space craft have to be judged according to the rules of the State of which the space craft has the nationality".[24]

On the other hand, it must be recognized that there are two other space law instruments dealing with special aspects of space activities, namely, the Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched into Outer Space of 1968 and the Convention on International Liability for Damage Caused by Space Objects of 1972. None of these treaties has stressed the need for defining outer space. The whole construction of the Liability Convention has been erected on the difference between the application of absolute liability to pay compensation for damage caused by space objects on the surface of the earth or to aircraft in flight, and the application of liability based on fault for damage caused elsewhere than on the surface of the earth to a space object or to persons or property on board such an object by a space object of another launching State. Nevertheless, the concept of outer space has penetrated in this Convention too, at least in an indirect way. For the term "space object" has to be interpreted and it will hardly be possible to do so without an exact content of the meaning of outer space.

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[24] A. Meyer, Legal Problems of Outer Space, in Vol. II. Space Explorations and Applications, Vienna, 14-27 August, 1968, at p. 1135 (United Nations, N.Y., 1969).

However, the fourth space law instrument, the Convention on the Registration of Objects Launched into Outer Space of 1975[25] has again pushed to the foreground the question of defining outer space, suggesting simultaneously an important criterion to be taken as the basis for a solution to this issue. According to Article II of this Convention the launching State is obliged to register its respective space object by means of an entry in an appropriate registry which it shall maintain, "when a space object is launched into orbit or beyond".[26] A similar, though less apparent approach has been applied in Article IV of the same Convention. In this provision States of registry, on the basis of practice recommended by the General Assembly Resolution 1721/XVI of 1961, assumed the obligation to furnish to the Secretary-General of the United Nations information on space objects carried on their registries. Such information must include basic orbital parameters, such as nodal period, apogee and perigee, of each object concerned.

In fact, the drafters of the Registration Convention applied the same criterion that had already been used by the designers of the 1967 Space Treaty in its Article IV dealing with the denuclearization of outer space. By this provision, in which an earlier commitment expressed in the General Assembly Resolution 1884/XVIII of October 17, 1963 was incorporated, States Parties to the Treaty have undertaken not to place in orbit around the earth any objects carrying nuclear weapons or any other kinds of weapons of mass destruction, not to install such weapons on celestial bodies or to station such weapons in outer space in any other manner.

As to the next space law documents the Draft Agreement Governing the Activities of States on the Moon and Other Celestial Bodies, as completed at the 1979 session of the UN Committee on Outer Space,[27] does not raise urgent questions with regard to the item of definition and/or delimitation of outer space. For its main aim is to regulate the exploration and use of the moon and other celestial bodies which lie, without any doubt, in outer space. On the other hand, two draft sets of principles which are still under discussion on remote sensing and on direct television broadcasting, use the terms of outer space and satellites ("remote sensing of the earth from outer space and international co-operation in that field"[28] and "activities in the field of international direct television broadcasting by means of artificial earth satellites").[29] In this way the definition of outer space, and particularly the element of satellite orbits, will again be invoked as it was in the case of the Registration Convention.

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[25]Convention on Registration of Objects Launched into Outer Space, March 29, 1972, [1973] 24 U.S.T. 2389, T.I.A.S. 7762 (effective Oct. 9, 1973).

[26]*Ibid.*

[27]*Cf.* Report of the Committee on the Peaceful Uses of Outer Space, General Assembly Official Records, Thirty-Fourth Session, Supplement No. 20 (A/34/20) Annex II, 33 (1979).

[28]*Cf.* Principle II in Texts of Draft Principles on remote sensing, U.N. Doc. A/AC.105/240 of April 10, 1979, Annex I, 8.

[29]*Cf.* Purposes and Objectives in Texts of Draft Principles on direct television broadcasting, U.N. Doc. A/AC.105/240 of April 10, 1979, Annex II, 9.

Therefore, it may be stated that at least in some principles and specific provisions of the relevant legal instruments, the criterion of orbits of earth satellites has already been applied, thus indicating a realistic and practical basis for a viable solution to the question of defining outer space. For all objects launched into orbit around the earth and beyond must be, without any hesitation, qualified as space objects, *i.e.,* objects accomplishing missions in outer space.

b) *Definition of Outer Space and New Generation of Space Vheicles*

The question of defining outer space has to be studied not only as a problem of the growth of space law on the basis of legal principles and rules as such, but also with due regard to new trends in the technology of space flights. Without any doubt, new generations of space vehicles will increase the present number of questions involved, adding further aspects to the existing issues and also raising some completely fresh points. The appearance of new types of space vehicle, as demonstrated by the Space Shuttle, offers an outstanding example of this development.

As we know, the Space Shuttle has to be launched as a spacecraft by means of rocket propulsion, but it will return to earth through airspace as a glider. This fact has inevitably brought up the question of whether the Shuttle, though obviously a space object during the initial stage of its flight and when fulfilling its space mission, may still be considered a space object when it derives in the final stage of its flight "support in the atmosphere from the reactions of the air", in the sense of the definition of aircraft as provided in Annex 7 to the Chicago Convention on International Civil Aviation of 1944.[30] Consequently, this double character of the Space Shuttle has been considered by some observers as a new major factor contributing to the practical significance of an agreement on a boundary between airspace and outer space.[31]

On the other hand, opponents of a conventional definition of outer space usually argue that the Space Shuttle will be registered as any other space object under the 1975 Convention on the Registration of Objects Launched into Outer Space and not as an aircraft under the 1944 Chicago Convention on Civil Aviation and the relevant law of an individual State governing aeronautics. Moreover, little difference is observed between the present space objects and the Space Shuttle from the point of view of descent profiles, since the Shuttle, too, "is operationally constrained to land at a pre-planned landing field".[32] Even if it is admitted that activities of the Space Shuttle should be

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[30]Chicago Convention on International Civil Aviation, Annex 7 (1944).

[31]Haanappel, Airspace, Outer Space and Mesospace, Proc. Nineteenth Colloquium on the Law of Outer Space 160 (1976).

[32]Menter, Relationship of Air & Space Law, Proc. Nineteenth Colloquium on the Law of Outer Space 166, 168 (1976). In this article reference is made to a letter of M.S. Malkin, Director of the Space Shuttle Program, NASA, addressed to M. Menter.

conducted "with due regard for the safety of other users of the airspace or for the safety of persons and property on the ground," it is still affirmed that this aim could be reached without defining the Shuttle as an aircraft. Thus, the whole character of the Space Shuttle is based on its purpose, "to engage in *space transportation rather than air transportation* as well as to engage in *space flight rather than aviation*. It is because of this that the "spacecraft nature" of the Shuttle predominates over its "aircraft nature".[33] The conclusion derived from this type of argument is that the appearance of the Space Shuttle has not in any way reinforced the need for the establishment of a delimitation between the two parts of space.

It should be noted, however, that notwithstanding that the Space Shuttle may be qualified as a space object for the purposes of registration, such qualification cannot change the fact that the Shuttle will move in two parts of space governed respectively by two different legal regimes. Consequently when considering practical questions of the co-existence of this type of space vehicle with other flying machines in the stage of traversing airspace, one cannot overlook the fact that it will move in this part of space in a similar way as other gliders, the only difference consisting in the origin and purposes of their flights.

Therefore the decisive question does not depend on whether the Space Shuttle in the descent stage of its flight shall be considered an aircraft or a spacecraft. When it enters that part of space which, if superjacent to the territory of a State, is subject to its complete and exclusive sovereignty, the Space Shuttle, too, as any other space object, has to observe the territorial supremacy of the State concerned including its national jurisdiction. The basic question, namely, at which moment of the flight such situation occurs, will always be valid and this question cannot be answered without a precise knowledge of the exact position of the boundary between airspace and outer space. Also sea vessels, when they navigate on the high seas, use the freedoms of the sea but when they enter the territorial sea of a State,they have to respect fully the sovereignty of the coastal State which extends beyond its land territory and internal waters up to the outer limit of the territorial sea. In a similar way, a space object of any kind, be it a traditional space object or a reusable aerospace vehicle, will be subject to the legal regime of outer space when in orbit. However, such object must observe the principle of sovereignty and acknowledge all legal consequences thereof when passing through the territorial airspace of a subjacent State.

Of course, it may be expected that trajectories of the Space Shuttle will be planned in such a way that besides the free part of airspace above the high seas they enter only the national airspace of the launching State. At the same time, however, eventualities of an unplanned landing should be taken into account. Moreover, it should be considered

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[33]Sloup, Why the NASA Space Shuttle Will Not Require a Specific Altitude to be Chosen As the Legal Boundary Between Air Space and Outer Space, Proc. Twentieth Colloquium on the Law of Outer Space 57 (1977).

that future aerospace vehicles may possess increased aerodynamic capability allowing them "to land and take off from any large and properly equipped airport on the planet".[34] The impact of a new generation of space vehicles cannot be studied only in the light of its first representative but also with due regard to other prospective types of aerospace transport vehicles.

Finally, the participation of other nations with less advantageous geographical positions, not allowing them to manage the flights of aerospace vehicles without violating some part of airspace subject to sovereignty of another State, should also be taken into consideration.

The only reasonable conclusion that might be derived from such situations would be a general recognition of the right of innocent passage through airspace above the territories of other States as and when necessary for the taking off and landing of a space vehicle, be it for purposes of a regular flight mission or when the vehicle concerned experiences emergency conditions. Such a solution would be primarily based on a analogy with the right of innocent passage through the territorial sea which must not be prejudicial to the peace, good order or security of the coastal State concerned and shall take place in conformity with relevant provisions of the law of the sea and other rules of international law. At the same time, however, we cannot neglect the differences existing between principles and other provisions of the law of the sea and the law governing space.

Moreover, it is possible to draw a certain analogy between the right of innocent passage through airspace and the right of access to and from the sea that is generally recognized in favor of the land-locked countries. Otherwise only a limited number of States would be in a position to use independently the freedom of outer space, while others might be in fact deprived of the right to explore and use outer space on an equal basis.

c) *Issue of Geostationary Orbit and Definition of Outer Space*

As already mentioned, the legal problem of defining outer space has been recently contaminated by another issue, namely the legal status of geostationary orbit. In contravention of para. 2 of Article I of the Space Treaty, which states that "outer space, including the moon and other celestial bodies, shall be free for exploration and use by all States, without discrimination of any kind, on a basis of equality and in accordance with international law", claims have been advanced to rights of sovereignty over certain segments of the geostationary orbit in favor of individual States located below such

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[34] Sloup, The Relationship of Air Law and Space Law—A View From the Space Shuttle, Including Its Internal and External Environments, Proc. Nineteenth Colloquium on the Law of Outer Space 210 (1976).

segments. The first statements of this kind were made as early as 1975 and 1976.[35] A group of equatorial States extended a similar position in a Declaration adopted at the meeting held in Bogota from 29 November to 4 December 1976.[36] It is evident that this document was drafted in connection with the World Administrative Radio Conference for the Planning of the Broadcasting Satellite Service in specific frequency bands, held in Geneva, 1977. Such claims were also presented and discussed in greater detail on the forum of the UN Committee on Outer Space and its Legal Subcommittee, particularly in 1977 and 1978.[37]

The advocates of claims of sovereignty endorsed them by different arguments, some of them even contradicting each other. In principle, however, their position was mostly based on the assumption that "the geostationary orbit, both because of its physical characteristics and technical attributes, but also because of the existing legal regulations, constitutes a limited natural resource over which the equatorial countries exercise sovereign rights in accordance with international law".[38] They also shared the opinion that

> "the provisions of international law which affirm the right of peoples and nations freely to exercise full and permanent sovereignty, including possession, use and disposition, over all their natural wealth and resources are applicable and justify the exercise of sovereignty over the segments of the geostationary orbit corresponding to their national territory".[39]

The delegations of the equatorial States in the UN Legal Subcommittee also maintained the viewpoint that "the geostationary orbit must be used in priority for the benefit of the developing countries in order to help to narrow the gap between the developing countries and the industrialized countries on an equitable basis."[40] Finally,

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[35]*Cf.* statement made on 14 October 1975 by the delegate of Colombia in the First committee of the U.N. General Assembly at its Thirtieth Session, U.N. Doc. A/C.1/PV.2052 at 43; statement of the delegate of Colombia made on 20 October 1976 in the First Committee of the U.N. General Assembly at its Thirty-First Session, U.N. Doc. A/C.1/31/PV.8 at 7. and statement of the delegate of Ecuador made on October 21, 1976 at the same forum, U.N. Doc. A/C.1/31/PV.10 at 36.

[36]Declaration of the First Meeting of Equatorial Countries (the so-called Bogota Declaration) was signed on December 3, 1976 by Colombia, Congo, Ecuador, Indonesia, Kenya, Uganda, Zaire and also by Brazil in the capacity of observer. For a text, see 6 J. Space L. 193 (1978). For a more detailed analysis of this document, *cf.* Finch, The Geostationary Oribt and 1967 Outer Space Treaty, Proc. Twentieth Colloquium on the Law of Outer Space 219 (1979).

[37]*Cf.* Gorbiel, Un nouveau probleme du droit cosmique international, 13 Revue roumaine d'etudes internationales 253 (2/44; 1979).

[38]*Cf.* Report of the Legal Subcommittee on the Work of Its Seventeenth Session (13 March-7 April 1978), U.N. Doc. A/AC. 105/218 of April 13, 1978 at 10. para. 40 (1978).

[39]*Ibid.*

[40]*Ibid.*

it must be recalled in the framework of this study that among arguments intended to endorse the sovereignty claims to segments of the geostationary orbit the lack of any satisfactory definition of outer space in the 1967 Space Treaty and other space law instruments was invoked, leaving reportedly to individual States to establish limits in the relevant parts of space by their own decision. Moreover, the equatorial countries stressed that with regard to the unique and specific nature of the geostationary orbit, "its *sui generis* character should be taken into account in any definition of outer space whose limits have not yet been established."[41]

As we know, representatives of other nations, including the socialist countries were resolutely opposed to such sovereignty claims and made this position abundantly clear both at the Geneva Conference of 1977 and in the United Nations. On several occasions, they emphasized that the geostationary orbit was an inseparable part of outer space, as a matter of fact the most important one in many aspects of the practical uses of present and future space technology. Like outer space as a whole, the geostationary orbit shall not be subject to any kind of national appropriation. As a part of outer space the geostationary orbit should be considered as the province of mankind, *i.e.*, it should be used for the benefit and in the interests of all countries, irrespective of the degree of their economic or scientific development and also irrespective of their geographical location. These and other reasons were advanced in the Working Paper submitted by the USSR to the UN Committee on Outer Space at its twentieth session in 1977, which contained considerations on the legal status of the geostationary orbit.[42] It should be also recalled in this connection that several analytical articles on this issue appeared in recent legal literature.[43]

The problem of the geostationary orbit must be definitely approached in quite another manner than in terms of national appropriation of its parts by individual countries, be it by claims of sovereignty or in fact by exclusive uses of the positions concerned and saturation of the geostationary orbit by a few nations. Both these ways

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[41] *Ibid.*

[42] Report of the Legal Subcommittee on the Work of Its Seventeenth Session, U.N. Doc. A/AC.105/L 94 at 10, para. 36 (1977); Report of the Committee on the Peaceful Uses of Outer Space. General Assembly Official Records, Thirty-Second Session, Supplement No. 20 (A/32/20), 1977, Annex VI on p. 29; Report of the Legal Subcomm. on the Work of Its Eighteenth Session, Mar. 12-Apr. 6, 1979, U.N. Doc. A/AC.105/240 at 9, 44, 47 (1979).

[43] *Cf.* Busak, The Geostationary Satellite Orbit—International Cooperation or National Sovereignty?, 45 Telecommunications J. 167 (1978); K. Wiewiorowska, Legal and Political Problems of the Geostationary Orbit, Proc. Twenty-First Colloquium on the Law of Outer Space 34 (1979); Christol, The Geostationary Orbit Position as a Natural Resource of the Space Environment, 26 Netherlands Int'l L. Rev. 5 (1979); Gorove, The Geostationary Orbit: Issues of Law and Policy, 73 Am. J. Int'l L. 443 (1979).

would lead to discrimination against nations of the present world even though they are all entitled to explore and use outer space as "the province of all mankind."[44]

On the contrary, the proper way to solve the problem would be to consider how to use this highly important part of outer space in the most rational and efficient way for the benefit of all countries, both those situated below this part of space and those located at other parts of our planet. A proper approach should also lead us to consider how to ensure equal rights to an effective participation in the utilization of the geostationary orbit for all nations interested in such activities, without creating obstacles to further useful developments of space exploration while serving the common interests of all mankind.

Definitely, the proper way to a settlement of this particular and fairly complex issue cannot be found in a division of outer space and in an isolation of its specific part that would be subject to completely different legal rules, neglecting and even opposing the fundamental principles of the space law in force as codified in the 1967 Space Treaty. On the contrary, on the basis of the general principles governing the exploration and peaceful uses of outer space as a whole, *i.e.*, under the general scope of the Space Treaty, some more specific rules regulating the uses of the geostationary satellite orbits could be developed, thus creating a special regime for this important but inseparable part of outer space within the general legal regime thereof.

As a matter of fact, despite the adoption of a somewhat misleading qualification of the geostationary orbit in Article 33 of the International Telecommunication Convention of 1973 (Malaga-Torremolinos) and other related documents[45], this method has already been applied in some practical measures by the ITU. Their purpose is to regulate "the best possible use of the radio-frequency spectrum and the

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[44]Art. I of the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies (hereinafter "Space Treaty"), Jan. 27, 1967, [1967] 18 U.S.T. 2410, T.I.A.S. No. 6347.

[45]*Cf.* Article 33 of the new version of the Convention entitled "Rational use of the radio frequency spectrum and of the geostationary satellite orbit" which specifies: "In using frequency bands for space radio services Members shall bear in mind that radio frequencies and the geostationary satellite orbit are limited natural resources, that they must be used efficiently and economically so that countries or groups of countries may have equitable access to both in conformity with the provisions of the Radio Regulations according to their needs and the technical facilities at their disposal." *Cf.* ITU, Thirteenth Report by the International Telecommunication Union on telecommunication and the peaceful uses of outer space 6 (Geneva, 1974).

In fact the geostationary orbit is not a resource as stated in this Article and some other ITU documents, but a functional position of an orbiting space object which maintains the same spot in outer space in relation to the earth and revolves round the earth with the same speed as our planet itself. Therefore, we should rather speak of geostationary satellite orbits the number of which is limited due to their unique position in space and use of radio frequencies. The phrase of "limited natural resources" in Article 33 of the International Telecommunication Convention should not be taken in the usual legal meaning of the term "natural resources" but understood rather as a comparison.

geostationary-satellite orbit as well as the need for an orderly development of the services to which these bands are allocated", simultaneously taking into account "the equal rights of all countries, large and small", even those countries which are not represented at the ITU actions.[46] The principle of preoccupation of all nations should remain how to develop further a reasonable and generally acceptable legal order governing present and future space activities in the interest of all countries, for the benefit of all mankind.

This would mean to take care of a genuine and continuing growth of the law of outer space. One of essential elements of the rule of law in outer space that should be settled now is just the delimitation of the exact territorial scope of validity of space law. Such delimitation cannot be accomplished without any precise definition of outer space. Had such definition, including an agreement on the lower boundary of outer space, been adopted in the earlier stage of space legislation, a firm legal obstacle against appropriation claims of individual States or any groups thereof would have been created and any claims of this kind could not have been advanced.

At the same time, it must be emphasized that the issue of the legal regime of the geostationary orbit cannot provide a proper answer to the question of defining outer space, both questions being of a different nature and significance. Even if we have to take into account all aspects involved, including the unique nature of the geostationary orbit, the question of defining outer space must be settled as such, *i.e.*, as an independent problem of the developing space law which is not to be subject to solutions of other issues.

III. *Proposed Solution to the Question*

If the definition of outer space is to serve the latter's purpose, it should be global and invariable, clear and simple, rapidly determinable both from the ground and outer space. Only when fulfilling such requirements would the definition of outer space be easily applicable, although in considering these criteria, physical and other scientific aspects of the problem must be taken into account. The substance of the need for defining outer space and delimiting it from airspace is primarily of a political and legal nature because, on the one hand, the sovereignty of States and the territorial scope of their laws and, on the other hand, the scope of the validity of international agreements governing the activities in outer space are at stake.

The need for a definition of outer space is a consequence of the development of space flights. If adopted, such definition has to further, and not to create legal obstacles to, the expected growth of space activities. For the aims of space flights the most important delimitation line is that which establishes the lower limit of outer space while the question of upper limit of "airspace", *i.e.*, of that part of space which is governed by

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[46] Cf. Preamble to the Final Acts of the World Administrative Radio Conference for the Planning of the Broadcasting-Satellite Service in Frequency Bands 11.7-12.2 GHz (In Regions 2 and 3) and 11.7-12.5 GHz (In Region 1) (Geneva, 1977).

the principle of complete and exclusive sovereignty of subjacent States (a limit which need not necessarily coincide with the above-mentioned lower limit of outer space) is not so urgent and might be left aside for the time being. The question of using this part of space for the purpose of transit of space objects might be settled in another way, *i.e.*, by a general recognition of the right of passage or, in other terms, the right of access of space objects to and from outer space, including the freedom of transit through the airspace of States other than the launching State. Of course, such right—as was correctly observed by Professor Manfred Lachs of Poland, Judge of the International Court of Justice—"does not connote activities which may be contrary to the principles and rules of outer space law or international law in general, or which are directed against a subjacent State or jeopardize its rights."[47]

Thus, the decisive criterion to be used as the basis for an agreement on establishing the lower limit of outer space should be of cosmonautical nature. In our opinion, there is but one such criterion that meets all the above mentioned requirements, namely, the lowest perigee at which space objects are still able to continue effectively in their orbiting around the earth for a longer period of time. Indeed, the lowest perigee of artificial earth satellites is and will remain constant for many years to come, notwithstanding the rapid progress in space technology. Moreover, such criterion meets the requirement of suitable applicability. As has been observed by Dr. Lubŏs Perek, an outstanding expert on scientific aspects of space, "the measuring of distance of any object in space can be made quickly with equipment which is not exceedingly expensive. Also the space objects themselves could, in principle, make such determinations."[48]

As to the exact height of the lowest perigee, it is possible to consult several studies published in recent years. In a thorough paper prepared by Working Group 1 of the Committee on Space Research upon the request of the U.N. Committee on Outer Space, entitled "Study on Altitudes of Artificial Earth Satellites", the conditions in the lowest altitudes at which satellites move and the disturbing forces, which can affect the diminishing of the height of the closest point of approach of a satellite to earth, were analyzed in great detail. According to this paper

> "it seems that the past estimates of the lowest heights into which satellites can plunge, without falling down to the ground or burning up in the atmosphere, were too high. This is especially true for satellites with highly eccentric orbits which penetrate into the atmosphere for a limited time during each revolution around the Earth."[49]

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[47]M. Lachs, The Law of Outer Space: An Experience in Contemporary Law-Making 61 (1972).

[48]L. Perek, Remarks on Scientific Criteria for the Definition of Outer Space, Proc. Nineteenth Colloquium on the Law of Outer Space 188-189 (1977).

[49]U.N. Doc. A/AC.105/164 at 4 and Annex I at 20 (1976).

As noted in this COSPAR study the lowest atmospherical depth into which artificial satellites of the earth have penetrated "is with good precision the height of 90 km."[50]

Therefore, the lower limit of outer space should be drawn not far from such height which represents a kind of natural boundary of space flights. Should the lower limit of outer space be located much higher, it would exclude from the realm of space law a significant part of space activities, including those with highly practical results. The performance of missions that would involve the penetration of objects into space below such limit would then require the consent of any subjacent State concerned. On the other hand, should the lower limit of outer space be located closer to earth, important interests of subjacent States falling within the category of national security might be affected without any proper benefit for space activities as such.

Furthermore, it is well to remember that other significant phenomena of a physical value also militate in favor of selecting the lowest relatively stable perigee of artificial earth satellites as the decisive criterion for the establishment of the lower limit of outer space. In the working paper entitled "Natural Boundaries in Space", submitted by Belgium at the thirteenth session of the Scientific and Technical Subcommittee of the UN Committee on Outer Space in 1976, the following facts were also recalled: The turbopause, being a boundary that separates two atmospheric regions with different physical properties, where air ceases to exist with its normal composition for principal constituents, is situated at the height of 100± km; the first persistent and important ionospheric level, *e.g.*, the so called E-region, occurs in the neighborhood of 100 km; atmospheric drag becomes important and also perceptible due to the glow of falling meteorites at 100± 10 km; important and vast atmospheric regions lie beyond this limit, such as plasmasphere, magnetosphere, etc.[51] Indeed, up to the height of 100 km the composition of atmosphere remains similar to that of its dense layer surrounding the earth. In more distant regions, its composition widely differs. All such coincidences, however, are but subsidiary arguments which support the validity of the main criterion on which the delimitation can be effectively based.

As a matter of fact the lowest perigee where space objects are still able to continue in their orbiting around the earth has already imposed itself as a natural criterion for the delimitation of outer space. This natural delimitation has also been evidenced by the constant practice of States and their tacit consent to space activities accomplished so far at this distance and beyond it. On the basis of such practice, Professor Wojciech Goralczyk of Warsaw University defends the view that a principle has already emerged whereby in the height of orbits around the Earth, where objects launched by States

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[50] *Ibid.*

[51] "The Question of the Definition and/or the Delimitation of Outer Space, U.N. Doc. A/AC.105/C.2/7/Add. 1 at 18-19 (1977).

move, the territorial supremacy of States does not prevail and such parts of space belong to outer space.[52]

Indeed, the criterion of the lowest perigee and the natural boundary of outer space arising from it should serve as the basis on which an agreement establishing a conventional boundary could be reached. Such conventional boundary might deviate from the basic criterion in one direction or the other taking into account all relevant viewpoints and interests of States. But in principle, it should follow the above mentioned natural line and confirm the already established custom.

In conclusion, it should be recalled that a Working Paper dealing with the approach to the solution of the problem was recently submitted by the delegation of the U.S.S.R. to the U.N. Legal Subcommittee on Outer Space. In this document it was suggested that an agreement be reached on the boundary between airspace and outer space at an altitude not exceeding 100/110 km above the sea level. At the same time, this document recommended retaining the right of space objects to fly over the territory of other States at lower altitudes for the purpose of reaching orbit or returning to earth in the territory of the launching State.[53]

We believe that this proposal has been well founded. It reflects actual practice of States performing space flights and at the same time preserves legitimate interests of subjacent States. Without any doubt, its adoption as the basis for a solution of the question of defining outer space would promote "the common interests of all mankind in the progress of the exploration and use of outer space for peaceful purposes".[54]

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[52]W. Goralczyk, Prawo miedzynarodowe publiczne w zarysie 236 (1979).

[53]U.N. Doc. A/AC.105/C.2/L. 121 (1979), *reproduced* in the Report of the Legal Subcommittee on the Work of Its Eighteenth Session, March 12-April 6, 1979, U.N. Doc. A/AC.105/240, Annex IV., at 6 (1979).

[54]*Supra* note 44.